Dynamic evolutionary change in post-Paleozoic echinoids and the importance of scale when interpreting changes in rates of evolution

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How ecological and morphological diversity accrues over geological time has been much debated by paleobiologists. Evidence from the fossil record suggests that many clades reach maximal diversity early in their evolutionary history, followed by a decline in evolutionary rates as ecological space fills or due to internal constraints. Here, we apply recently developed methods for estimating rates of morphological evolution during the post-Paleozoic history of a major invertebrate clade, the Echinoidea. Contrary to expectation, rates of evolution were lowest during the initial phase of diversification following the Permo-Triassic mass extinction and increased over time. Furthermore, although several subclades show high initial rates and net decreases in rates of evolution, consistent with "early bursts" of morphological diversification, at more inclusive taxonomic levels, these bursts appear as episodic peaks. Peak rates coincided with major shifts in ecological morphology, primarily associated with innovations in feeding strategies. Despite having similar numbers of species in today's oceans, regular echinoids have accrued far less morphological diversity than irregular echinoids due to lower intrinsic rates of morphological evolution and less morphological innovation, the latter indicative of constrained or bounded evolution. These results indicate that rates of evolution are extremely heterogenous through time and their interpretation depends on the temporal and taxonomic scale of analysis.

fossil record | morphological diversification | early bursts | evolutionary innovation | mode of evolution

Assessing how rates of morphological evolution have changed over geological time has been a major research goal of evolutionary paleobiologists since Westoll's classic study of lungfish evolution (1). A common pattern to emerge from the fossil record is that many clades reach maximal morphological diversity early in their evolutionary history (2–4). This sort of pattern could be the result of an "early burst" of morphological diversification as taxa diverge followed by a slow-down in rates as ecological space becomes filled (5, 6). Internal constraint or long-term selective pressures could also limit overall disparity, leading to a slowdown in the rate of new trait acquisition over time (7, 8). However, only a small proportion of fossil disparity studies have also assessed changes in rates of evolution within lineages (e.g., along phylogenetic branches) thereby providing a more nuanced understanding of how this disparity came about (e.g., refs. 9–13). Simultaneously, decreasing rates in trait evolution have been difficult to detect using phylogenetic comparative data of extant taxa, because of low statistical power (14, 15), loss of signal through extinction (16), and inaccuracies in reconstructing ancestral nodes (17). Here we take advantage of recently developed methods for directly estimating per-lineage-million-year rates of evolution from phylogenies with both fossil and living taxa to test whether declining rates characterize the evolutionary history of a major clade of marine invertebrates, the echinoids.

Since originating some 265 million years ago (18, 19), crown group echinoids have evolved to become ecologically and morphologically diverse in today's oceans, and are an important component of both past and present marine ecosystems (e.g., refs. 20–22). However, analysis of how this diversity arose has either been based on taxonomic counts (e.g., ref. 23) or has adopted a morphometric approach where the requirement of a homologous set of landmarks limits taxonomic, temporal, and geographic scope (e.g., ref. 24). We use a discrete-character-based approach and a recent taxonomically comprehensive analysis of post-Paleozoic echinoids as our phylogenetic framework (25). This tree is almost entirely resolved (*SI Appendix*, Fig. S1) and branches may be scaled using the first appearance of each taxon in the fossil record (*SI Appendix*, Table S1). We tabulated the number of character state changes that occurred along each branch within ∼10-million-year time intervals spanning the Permian and post-Paleozoic (*SI Appendix*, Table S2), and divided this by the summed duration of branch lengths to compute a time series of per-lineage-million-year rates of morphological evolution. We accounted for uncertainty in phylogenetic structure, uncertainty in the timing of the first appearance of taxa, and uncertainty in the timing of character changes along each branch using a randomization approach (12). We also estimated rates within subclades, corroborating our findings by using likelihoods tests to determine whether some branches had higher rates than expected given rates across the entire tree. Finally, we compared rates of evolution through time with the structure of diversification within a character-defined morphospace, and looked for evidence of differences in evolutionary modes among subclades. The pattern that emerges is one of dynamic evolutionary change through time: Both rates and patterns of evolution vary

Significance

Biodiversification studies have often relied on constant-rate models of diversification. More recently, however, there has been an effort to identify changes in diversification rates within clades. This effort has largely focused on models of declining rates because many clades appear to have high initial rates, followed by slow-downs as ecological space fills. Here we provide an example of a 265 million-year-old marine invertebrate clade where evolutionary rates show a net increase over time instead. This is punctuated by intervals of high rates of morphological evolution, coinciding with major shifts in lifestyle and the evolution of new subclades. This study demonstrates the dynamic nature of evolutionary change within major clades.

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temporally and across subclades, such that the overall pattern depends highly on the temporal and taxonomic scale of the analysis.

Results and Discussion

For crown-group echinoids as a whole, rates of character change fluctuated throughout the post-Paleozoic, but there was a net increase in rates overall (Fig. 1, Spearman's *Rho* = 0.47, $P = 0.017$). In addition, rates more than doubled at the start of the Jurassic, exceeding preceding time intervals. Rates remained high for the next 40 million years, dropping dramatically at the end of the Jurassic to a rate slightly higher but still within the range of rates during the Triassic. Rates did not increase again until the Early Eocene, where they again reached levels achieved in the Jurassic, and remained high for about 20 million years. Although they fall again after the Eocene, rates are elevated to the present day compared with rates estimated for the Triassic, and have remained relatively constant since the Oligocene. Rates were at their lowest earlier in the clade's history, particularly during the initial recovery (and putative ecological release experienced by surviving lineages) following the Permo-Triassic mass extinction. The secular increase in average rates of character change is even more apparent if the episodic peaks (rates > 0.24) are ignored (Spearman's *Rho* = 0.79, $P < 0.001$).

These results are robust to variation in the age data, character optimization, time-scale resolution, and tree-scaling methods (*Methods* and *SI Appendix*). The pattern is also surprisingly robust to the exclusion of fossil taxa as tips (*SI Appendix*, Figs. S4 and S5), although how far back in time rates can be inferred depends on whether completely extinct taxa are used to scale the tree or not (*Methods*). Thus, the exclusion of morphological changes along branches leading to now-extinct lineages did not have a noticeable effect on the pattern of inferred rates of evolution through time, indicating that most changes are captured along the branches of the pruned tree leading up to them, and that past morphological diversity is encompassed by modern echinoids. We do not suggest, however, that this would be true for other clades. Extinction in crown-group echinoids has largely been limited to intermediate forms linking major subclades (see below). In clades where significant or outlying morphological diversity has been lost through extinction, the exclusion of fossil taxa as tips could lead to very different inferred rates of evolution.

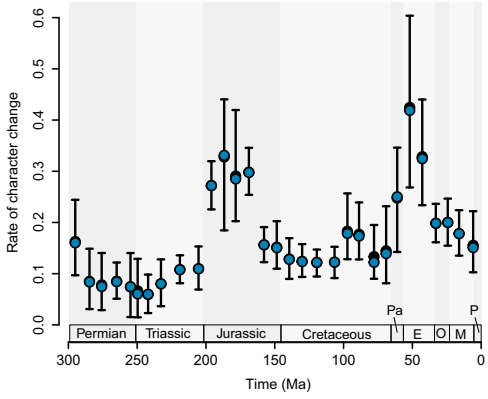


Fig. 1. Rates of morphological evolution in echinoids since the Paleozoic. Rates are measured as the mean number of character changes per lineage million years within ~10-million-year time intervals, accounting for uncertainty in phylogenetic structure, timing of first appearances of taxa in the fossil record, and uncertainty in timing of character state changes along branches. Error bars represent 95% confidence intervals. Black points = mean rate values; blue = median rate values. For most time intervals, mean and median rates values are the same. E, Eocene; M, Miocene; Ma, millions of years ago; O, Oligocene; P, Plio-Pleistocene; Pa, Paleocene.

Sampling biases could make rates of evolution appear elevated during the Jurassic and Eocene when they have actually been constant through time. For example, a prior interval of poor preservation or under sampling may fail to capture a significant portion of taxa that originated during that period, these taxa making their first appearance in the fossil record coincidently only when preservation potential improves, thus creating an artificial spike in origination. We assessed changes in the quality of the echinoid fossil record by dividing the number of lineages sampled within each time interval by the number of lineages inferred to be present from the phylogenetic analysis (*SI Appendix*, Table S3). We found a nonsignificant correlation between changes in this measure of completeness and changes in estimated rates of character change, (Spearman's *Rho* = −0.24, $P = 0.25$, *SI Appendix*, Fig. S6). We also estimated sampling intensity using the number of collections that included echinoid taxa sampled globally from each time interval (*SI Appendix*, Table S4), and again found a nonsignificant correlation between changes in sampling intensity and changes in estimated rates of character change (Spearman's *Rho* = −0.29, $P = 0.153$, *SI Appendix*, Fig. S6). In both cases, there is also no support for strong lagged associations between the variables, and the results are robust to time-scaling methods (*SI Appendix*, Table S5), indicating that rate changes were not due to sampling artifacts biasing first appearances estimates.

Incomplete sampling could also result in completely missing branches. Missing branches will result in a decrease of the number of character changes and in the summed branch length estimated for any given time interval, so the effect on estimated rates will depend on the proportional decrease of each. Rates will be artificially depressed by the absence of taxa if those taxa have otherwise unrepresented morphological disparity. However, we have sampled 94% of the crown-group echinoid families, and the taxa omitted are not unusual in their morphology, just poorly documented, so we do not believe that we are missing any morphological outliers.

Although numerous events in Earth history could have influenced rates of morphological evolution in echinoids, both periods of elevated rates appear to be closely associated with major eco-morphological shifts. Increased rates in the Jurassic coincided with the early expansion of irregular echinoids as they adapted to deposit feeding for the first time in the clade's history (26, 27) (Fig. 2*A*). In addition, the first appearance of the grazing trace fossil *Gnathichnus* in the Rhaetian (28) indicates that regular* carinacean echinoids had started to rasp hard substrates by this time (Fig. 2*C*). This ecological shift to graze-feeding required the evolution of denser tube feet and thus pore-pair crowding, necessitating the adoption of plate compounding. Similarly, the increased rate of morphological evolution during the Eocene coincided the adoption of a gravity-assisted sieving feeding strategy in crown group clypeasteroids (Fig. 2*B*), a strategy long associated directly and/or indirectly with several morphological innovations, including differentiation of spines, test flattening, the evolution of lunular perforations through the test, and a branched food-groove system by which particles are efficiently passed to the mouth (29). However, clypeasteroids were also specializing for life in high-energy shallow-water environments at the time (30). To assess the relative support for these two hypotheses, we scored the 51 character changes that occurred

*Here we use the term "regular" to refer to all echinoids within the Cidaroidea and Euechinoidea minus the Irregularia. Although regular echinoids comprise a paraphyletic group, it is useful to be able to refer to them together because they share morphological and ecological characteristics, most notably pentaradial symmetry, aborally located periproct, and an exclusively epifaunal habitat. The Irregularia comprise a monophyletic group nested within the Euechinoidea, and are characterized by bilateral symmetry, periproct opening outside the apical disc, and are almost all semiinfaunal or infaunal deposit feeders.

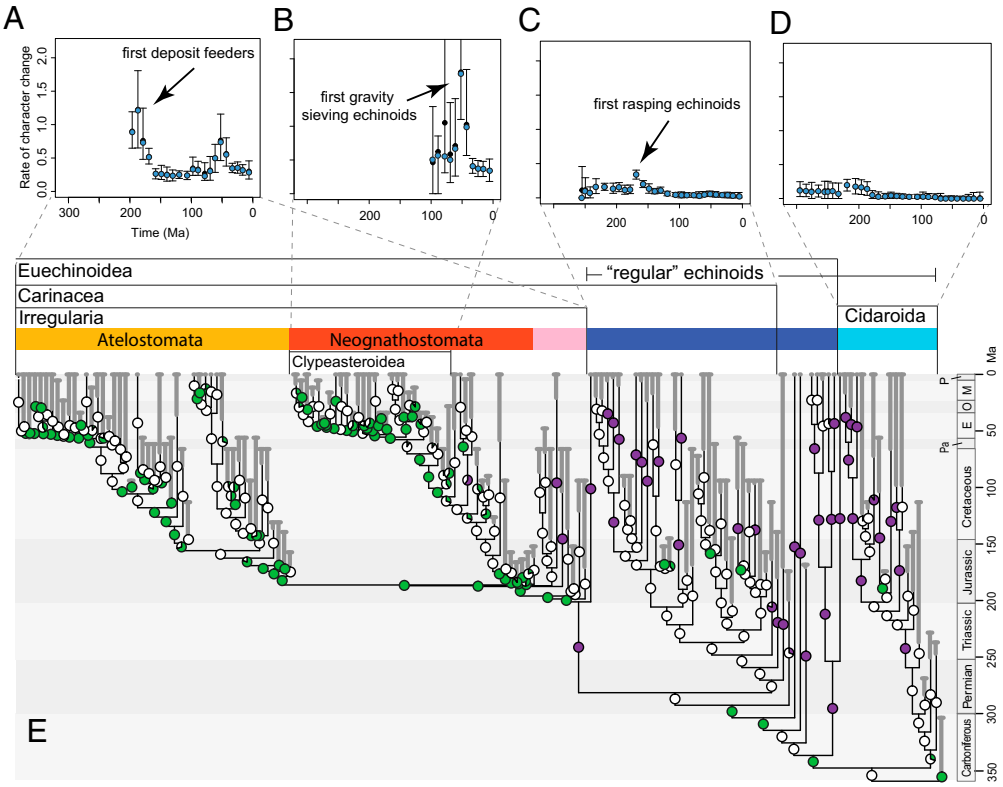


Fig. 2. Rates of morphological evolution in different echinoid subclades: Irregularia (*A*); clypeasteriods (*B*); regular carinacean echinoids (*C*); and cidaroids (*D*). All data are plotted on a similar axial scale for easy comparison at the expense of some error bars early in the history of the clypeasteroids (for *B*–*D*, see X and Y axes in *A*). (*E*) Results of branch likelihood ratio test (12) shown on scaled phylogenetic tree with randomly resolved local polytomies. Ratio test was repeated iteratively to account for uncertainty in timing of first appearances; pie charts show percentage of tests where branch rates were significantly different (either higher in green, or lower in purple) from the rest of the tree. Gray bars show stratigraphic range of each genus. Tip labels excluded for clarity, but see *SI Appendix*, Fig. S1. Horizontal colored bars indicate subclades plotted in Figs. 3 and 4. Ma, millions of years ago.

during the early evolution of clypeasteroids as related to (*i*) feeding, (*ii*) hydrodynamics, or (*iii*) neutral with respect to feeding or hydrodynamics. We found that a much larger proportion of nonneutral characters were related to feeding (74%) than hydrodynamics (26%) (*SI Appendix*, Table S6).

Notably, rates of character change within some subclades decrease over time (Fig. 2 *A* and *B* and *SI Appendix*, Fig. S7), consistent with "early bursts" of morphological diversification. These bursts are nested within the evolutionary history of more inclusive clades. For example, clypeasteroids show an early burst of morphological diversification in the Eocene that declines in younger time intervals (Fig. 2*C*). This burst contributes to the peak in diversification seen in the Eocene in Irregularia as a whole (Fig. 2*A*), whereas the burst of morphological diversification that characterizes the early evolution of this larger subclade contributes to the peak in diversification seen in crown-group echinoids as a whole during the Jurassic (Fig. 1). Thus, the apparent pattern of change in evolutionary rates is dependent on the taxonomic scale: At increasingly inclusive taxonomic levels, early bursts appear as episodic peaks in the overall diversification history.

The net increase in rates in post-Paleozoic echinoids overall is not simply due to the addition of younger subclades that may not have had time for rates of evolution to decline substantially (14), but due to higher intrinsic rates in irregular echinoids than in regular echinoids, particularly cidaroids (Fig. 2 *A*–*D* and *SI Appendix*, Fig. S7). This finding is corroborated by likelihood ratio tests, showing that most branches within the Irregularia part of the tree show significantly faster rates of morphological evolution than the rest of the tree, and many branches within the regular echinoid part of the tree show significantly slower rates of evolution than the rest of the tree (Fig. 2*E*). The difference in rates between regular and irregular echinoids is surprising given the fact that current species diversity in both is roughly the same (31). However, the taxonomic diversification history of echinoids does not mirror the morphological diversification history: For example, large numbers of new genera appear during the Cretaceous and Paleocene when rates of character change were relatively low (*SI Appendix*, Fig. S8). This decoupling suggests that elevated evolvability does not always elevate origination rates (cf. 32).

The difference in rates between irregular and regular echinoids may be inflated by sampling differences between the two. Irregular echinoids have a better fossil record: This can be seen in the greater number of unsampled ghost ranges in regular echinoids (49%) vs. irregular echinoids (24%) (*SI Appendix*, Table S3) and in the greater number of collections containing irregular echinoids in most time intervals (*SI Appendix*, Table S4). Despite this difference, sampling intensity is not significantly correlated with rates of character change within irregular echinoids (Spearman's $Rho = -0.08$, $P = 0.766$), within regular carinaceans (Spearman's $Rho = -0.02$, $P = 0.951$) or within cidaroids (Spearman's $Rho = -0.26$, $P = 0.200$). Generally, there is considerable variability in sampling intensity of both regular and irregular echinoids throughout the post-Paleozoic, but episodic peaks in irregular echinoids occurred much less frequently than large shifts in sampling intensity and rates within regular echinoids remained relatively steady. Even when sampling intensity is greater in regular echinoids than irregular echinoids (the late Jurassic), rates remain lower in the former than in the latter. Thus, the difference in rates among subclades cannot be

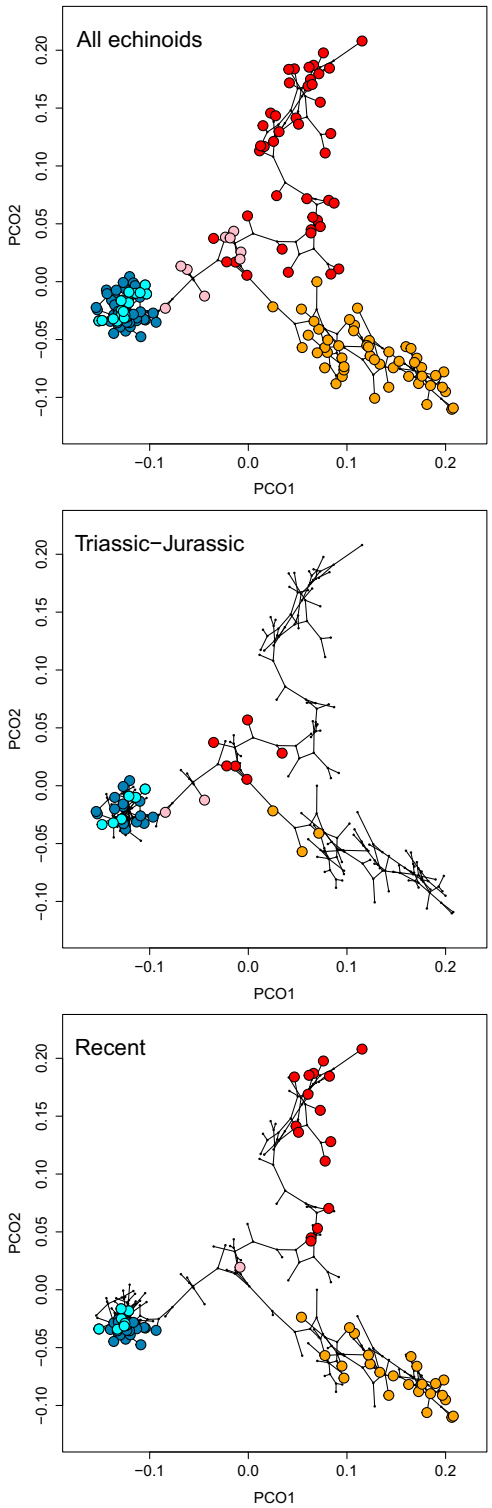


Fig. 3. Phylomorphospace based on principal coordinates analysis of character matrix, showing morphological disparity of post-Paleozoic echinoids realized throughout the entire history of the clade (*Top*), during the Triassic and Jurassic (*Middle*), and among living echinoids (*Bottom*). Warm colors indicate irregular echinoids: red, Neognathostomata; orange, Atelostomata; pink, stem groups. Cool colors indicate all regular echinoids: cyan, Cidaroida; blue, regular euechinoids. The first two principal coordinates together summarize 57.3% of the variation.

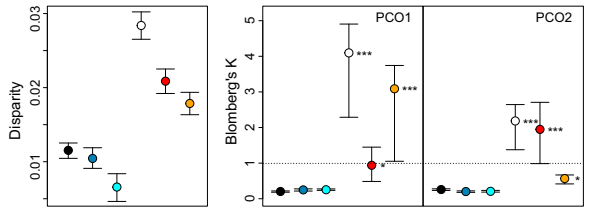


Fig. 4. Disparity (*Left*) and phylogenetic signal (*Right*) in irregular vs. regular echinoids. Error bars represent 95% confidence intervals. White, all irregular echinoids; black, all regular echinoids; otherwise, colors as in Figs. 2 and 3. Asterisks indicate *P* values for significant phylogenetic signal based on the permutation tests: $*P < 0.01$ for all trees; $***P < 0.001$ for all trees.

due solely to differences in the quality of the fossil record of each.

There is also a remarkable difference in overall morphological disparity within regular echinoids compared with irregular echinoids (Figs. 3 and 4). Living regular echinoids occupy the same region of character-based morphospace as Triassic and Jurassic regular echinoids (Fig. 3), indicating that they largely share the same suite of characters as their ancestors, and have not diversified much. In contrast, living irregular echinoids are considerably more morphologically diverse than their ancestors (Fig. 3) and shifted their morphospace occupation through time (*SI Appendix*, Fig. S9), indicting continued acquisition of new suites of characters through time.

These differences in new character acquisition over time suggest that irregular and regular echinoids show differences in evolutionary mode as well as evolutionary rate. Conceptually, this is similar to making a distinction between magnitude of change along branches and direction of changes along branches (33). To explore this further, we calculated phylogenetic signal in the PCO1 and PCO2 scores shown in Fig. 3. Previous work has shown that phylogenetic signal is eroded in situations where the rate of evolution is fast relative to the boundaries of occupied phenotypic space (i.e., where morphological evolution is constrained), whereas early burst scenarios should preserve or even increase phylogenetic signal (34). As expected, phylogenetic signal in regular echinoids is much lower than that in irregular echinoids (Fig. 4). *K* values so much greater than one in irregular echinoids may reflect decreasing rates over time (Fig. 2*A* and *SI Appendix*, Fig. S7). However, these high values may also be due to the fact that evolution within subclades of irregular echinoids has been directed in morphospace, so closely related taxa resemble one another more than expected under Brownian motion. In particular, atelostomates show more divergence along PCO1 than PCO2, whereas neognathostomates show more divergence along PCO2 than PCO1; in both cases, the different *K* values reflect this.

Thus, irregular echinoids comprise more morphological diversity both because they have faster per-branch rates of character change and because those character changes have more frequently involved the innovation of new traits, whereas regular echinoids have evolved more slowly and have been more constrained in the directions that morphological evolution has taken. In addition, the disparity between regular echinoids and the two major clades of irregular echinoid (Atelostomata and Neognathostomata) has increased through time. This increase has been driven by the expansion into new areas of morphospace by the two irregular echinoid clades, and by elimination via extinction of the great majority of intermediate forms linking the irregular and regular echinoids as well as those linking Atelostomata and Neognathostomata. We hypothesize that higher intrinsic rates in irregular echinoids are associated with an important switch in growth strategy: Plate addition became much less important than plate accretion for test growth in irregular echinoids and led to the

early ontogenetic fixing of plates forming lower and upper test surfaces, facilitating regional differentiation of test morphology (35).

This study demonstrates the importance of the scale of analysis when discussing rates of evolution. At the largest taxonomic and temporal scales, as the echinoid skeleton has increased in complexity over time (as measured by number of scoreable character states) so too has the rate of character change. However, this has been achieved through a series of localized bursts of innovation within subclades as breakthroughs led to new ecologies being adopted. Because these bursts were episodic, morphological diversification across all post-Paleozoic echinoids continued long after the first crown-group echinoids evolved. In fact, maximum morphospace occupation was attained more than 200 million years after the origination of the earliest crown-group members, and may increase again sometime in the future. Both "continuous" and "early burst" patterns of diversification apply within the same clade, depending upon the scale of the analysis, and represent members of a spectrum of rate changes. Establishing the shape of the spectrum will be important for the accurate dating of node divergence times using fossils. We believe this is likely to be the case for most major clades, but this remains to be tested.

Methods

Phylogenetic Analysis. We used the strict consensus tree of post-Paleozoic echinoids presented in (25) to infer rates of character change. The taxa included in this tree are primarily type species representing 164 of 174 families of crown group echinoids, and over half are fossil taxa. This tree is almost entirely resolved with only a small number of local polytomies (*SI Appendix*, Fig. S1). Because the method that we use for inferring rates (*Rates of Character Change*) requires a fully resolved tree, we randomly resolved the consensus tree 100 times and then mapped character changes onto each new topology using both accelerated and delayed transformation (ACCTRAN and DELTRAN, respectively, ref. 36), each of which approximates an end-member of the full spectrum of optimization values within a parsimony framework. As noted by Lloyd et al. (12), it could be informative to vary the optimization character-by-character, but this approach is so computationally demanding as to be impractical. In addition, results are robust to choice of optimization (*SI Appendix*, Fig. S2*B*).

Because the analysis includes fossil taxa, the dataset was necessarily limited to features of the highly preservable exoskeleton. Fortunately, these features can frequently be linked to aspects of the nonmineralized phenotype, and the relationship between these features with the development, ecology, and physiology of fossil echinoids may be made by analogy with living representatives from these groups (37).

Rates of Character Change. Most methods currently available for detecting shifts in diversification rates along branches are limited in their application to ultrametric trees, necessarily limiting information to contemporaneous (typically Recent) taxa outside of the use of older taxa for dating nodes (e.g., refs. 38–40). Instead, we estimated rates of character change per time interval following Lloyd et al. (12). This method uses a randomization approach to explicitly account for two sources of uncertainty: (*i*) timing within the geologic stage of the first appearance of each taxon; and (*ii*) timing of character transformations along each branch. Because genus stratigraphic ranges are known with more accuracy in the fossil record than species stratigraphic ranges, we used the first appearance of the genus to which each species belonged to scale the branch lengths of the tree. Numerical dates for the first appearance of each taxon were assigned by drawing at random from a uniform distribution bounded by the maximum and minimum ages of the geologic stage in which the taxon first appeared (*SI Appendix*, Table S1). Using these ages, we scaled the tree by assigning each internal node the age of its oldest descendent, and then modifying zero-length branches so that each shares time equally with a preceding, non-zero-length branch (24). We assigned numerical dates to each character state transformation by drawing at random from a uniform distribution bounded by the start and end dates of the branch on which the state change occurred, as inferred from the scaled phylogenetic tree. We then divided the sum of changes by the summed duration of the branches within ∼10-million-year time intervals (*SI Appendix*, Table S2) to get a per-lineage-million-years rate of character change. We repeated this procedure 100 times for each resolved tree, for a total of 10,000 time series, and combined the results to assess the sensitivity of our rate estimates to uncertainty in the timing of first appearances, occurrence of character transformation along branches, and phylogenetic structure. We chose this number of iterations because larger numbers of time series did not change the results (*SI Appendix*, Fig. S2*A*), but significantly increased processing time. We ran all analyses in R (41), using script published in ref. 12 and functions available from the paleotree R library (42).

Results are robust to different aspects of the input data or methods, including: (*i*) character optimization method (*SI Appendix*, Fig. S2*B*); (*ii*) using family-level first appearances instead of genus-level first appearances to scale the tree (*SI Appendix*, Fig. S2*C* and Table S1); and (*iii*) using a higher-resolution time scale where time intervals average 5 million years instead of 10 million years (*SI Appendix*, Table S7 and Fig. S2*D*). In the latter case, the larger confidence intervals likely reflect the fact that the average branch length (27 million years) is greater than the average time interval in both low- and high-resolution time scales (10 million years and 5 million years, respectively), and more so for the high-resolution timescale. Of the variations applied, the results are most sensitive to the method of time-scaling branch lengths (*SI Appendix*), particularly in the size of confidence intervals around rates estimated from the early history of the clade (*SI Appendix*, Fig. S3). Nonetheless, the overall pattern is largely the same with rates peaking in the Jurassic and Eocene.

It is important to note that the scaling of the deepest nodes is highly sensitive to the choice of outgroup. *Archaeocidaris* was chosen as the outgroup in the original phylogenetic analysis because it is an unambiguous late stem-group member and its morphology is well documented (25). However, there is as yet no evidence that it is the closest true sister taxon to the crown group. In addition, its first appearance was during the Tournaisian (358.9–346.7 Mya), about 50 million years before the first appearance of the oldest crown group member in the analysis (*Eotiaris*), and the scaling of the deepest nodes of the tree is spread out across this time interval. A younger outgroup would have compressed these branches, possibly increasing inferred rates of evolution in the Permian. A different outgroup would also have resulted in a different number of character changes along the branch separating the outgroup from the rest of the clade. Because of this sensitivity to the choice of outgroup, we focused on the evolutionary history of echinoids since the Permian. All tests of association are based on rates calculated from the Triassic to the Recent.

When we pruned the tree to include only living taxa and then scaled it using the first appearance of those taxa as observed in the fossil record (*SI Appendix*, Figs. S4 and S5*A*), we were no longer able to infer rates for echinoids before the Jurassic (no living echinoid genera are older than mid-Jurassic) but the rates were noticeably higher in the late Jurassic and somewhat so in the Eocene than in the rest of the time series. However, scaling the tree based only on the fossil record of living taxa imparts unreasonably young ages to deep nodes, compared with both the known fossil record and to estimates inferred from molecular phylogenies of echinoids using current methods for calibrating trees and at least some information from the fossil record to define priors (18, 19). If we pruned the tree after scaling (imparting more reasonable ages to ancestral nodes), the pattern remains remarkably similar (*SI Appendix*, Fig. S5*B*). In both cases, branch tips were dated at the present.

To extract rates of morphological evolution within clades, we reran the entire analysis with trees pruned to the following subsets of taxa: (1) all regular echinoids; (2) Cidaroida; (3) regular Carinacea [excludes Irregularia]; (4) Irregularia; (5) Clypeasteroida; (6) nonclypeasteroid irregular echinoids; (7) Atelostomata; (8) Neognathostomata. Groups 1, 3, and 6 are paraphyletic but share morphological and ecological characteristics. We pruned trees after randomly resolving polytomies, optimizing character transformations along branches, and time-scaling the tree. In this way, counts of character changes were based on the same tree(s) as the full analysis, but made along subsets of branches. This procedure was computationally less intensive than keeping track of the affinity of each character change at different taxonomic levels and along each tree branch in the full analysis.

See *SI Appendix* for further details on the time series analyses for sampling bias.

Branch Likelihood Test. To determine whether some branches had higher or lower rates of character change than expected, we used a likelihood-based approach described in detail in Lloyd et al. (12), and briefly in the *SI Appendix*. We applied the likelihood ratio test 1,000 times to one tree with randomly resolved polytomies. For each iteration, we (re)scaled the tree in the same manner as described above to account for uncertainty in branch lengths resulting from uncertainty in the timing of taxon first appearances. Pie charts along each branch in Fig. 2*E* show the proportion of iterations that

that branch was found to have either a significantly faster or slower rate compared with the rest of tree.

Principal Coordinates Analysis. We generated a morphospace for post-Paleozoic echinoids by running a principal coordinates analysis (43, 44) of the character matrix used to create the phylogeny (ref. 25, appendix 2), after correcting a few errors in the published matrix (*SI Appendix*). The principal coordinates analysis is an ordination of dissimilarities among taxa, the purpose of which is to summarize as much of the variation between specimens as possible on as few axes as possible; it is synonymous with classical or metric multidimensional scaling (44). We calculated dissimilarity between pairs of taxa using Gower's coefficient (45). Briefly, if two taxa share the same character state for a character, they are assigned a value of 0 for that character, otherwise a value of 1. The number of mismatches (in character states) is divided by the total number of applicable or nonmissing characters (see *SI Appendix* for more detail).

We estimated the amount of variation summarized by each principal coordinate axis by dividing its eigenvalue by the sum of all positive eigenvalues (44). We estimated the total disparity within each echinoid group by summing the variance along each positive eigenvector; 95% confidence intervals are based on bootstrapping with replacement (1,000 times). We mapped the phylogenetic history onto the morphospace using the phytools package in R (46); in this implementation, the principal coordinate scores of the ancestral nodes are estimated using a maximum likelihood algorithm (33, 47). However, we use this only to illustrate the phylogenetic structure between taxa, and do not attempt to draw conclusions about the morphology at those nodes.

Phylogenetic Signal. To assess phylogenetic signal, we treated the scores for PCO1 and PCO2 as two continuous variables associated with the tree tips and then calculated Blomberg's K (48) for these variables using the phylosignal function in the picante R package (49). We calculated Blomberg's K, along with the associated permutations test (1,000 reps), for 10,000 trees with randomly resolved polytomies, scaled branch lengths, and pruned as described above for calculating rates of evolution within subclades. Under a Brownian motion model, Blomberg's K has an expected value of 1.0; when $K > 1.0$, closely related tips resemble one another more than expected under a Brownian motion model, when $K < 1.0$, closely related tips resemble each other less than expected (phylogenetic signal is low). Although there are other metrics for phylogenetic signal, we chose to use Blomberg's K because it is not limited to values below 1.0 (cf. Pagel's lambda, ref. 50) and because its behavior under different evolutionary scenarios has been previously documented (34).

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